Homework Assignment – Mario 2D Platformer in Unity

Objectives

Create a basic 2D platformer game in Unity using the provided Mario character, sprites, and sound effects. This assignment will help you practice importing assets, setting up a player controller, handling collisions, and building a simple game loop.

Requirements

1. Player Character (Mario)

~~-~~ Import the Mario model (provided).
~~-~~ Use Rigidbody for movement.
~~-~~ Implement:
- Move left/right for forward/backward with arrow keys or WASD.
- Flip player face according to movement direction.
- Jump with Spacebar (use Unity's Physics.gravity).

~~-~~ Add animations (walk, idle, jump).
~~-~~ Play a jump sound when Mario jumps.

2. Environment

~~-~~ Create a ground plane or terrain.
~~-~~ Add platforms (textured blocks) at different heights.
~~-~~ Use textures from the provided Mario sprites (bricks, grass, etc.).

3. Collectibles (Coins)

~~-~~ Place 5 coins in the level.
~~-~~ When Mario touches a coin:
- Destroy the coin (Destroy(gameObject)).
- Play a coin sound.
- Increase score by +1.

4. Obstacles / Enemies

~~-~~ Add at least 1 enemy (e.g., a beetle).
~~-~~ If Mario collides with the enemy:
- Reduce lives by one (start with 3).
- Play a "hit" sound.

~~-~~ If Mario jumps on top of the enemy → enemy is destroyed.

5. UI

- Add a Canvas with:
- ~~⊖~~ • Score text (coins collected).
- ~~⊖~~ • Lives text (remaining lives).

6. Level Goal

- ~~⊖~~- Add a flagpole or castle prefab at the end of the level.
- ~~⊖~~- When Mario touches it:
  - Show “Level Complete!” in the UI.
  - Stop player movement.

Deliverables

1. Zipped Unity project folder (exclude packages, bin, obj etc).
2. Gameplay video (1–2 minutes).
3. A readme.txt including:
   - Controls used
   - Features implemented
   - Any known bugs

4. This checklist with Check mark what is completed.